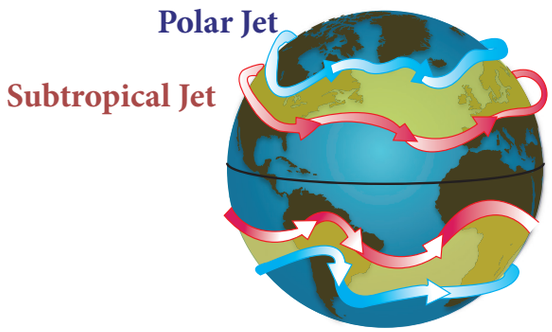



**Figure 6.19** Jet Stream

**The Major impacts of Jet streams**

1. **Creation of Polar Vortex:** Polar westerly jet stream will carry cold polar air masses towards temperate region which creates severe cold waves in North America and Eurasia during winter.
2. **Sudden burst of South west monsoon:** Sudden withdrawal of polar westerly jet stream from Indian sub continent to northern part of Pamir, leads to sudden burst of South west monsoon into Indian Sub continent.
3. **Late and early monsoon in South Asia:** Rate of with drawl of polar westerly jet stream decides the onset of south west monsoon. Slower and faster rate of with drawl leads to late and early onset of south west monsoon.
4. **Intensity of monsoon rainfall:** The arrival of tropical easterly jet stream influences the intensity of south west monsoon. This leads to increasing intensity of rainfall during south west monsoon.

**DO YOU KNOW?**

Jet streams were discovered during the Second World War as the jet pilots felt the strong obstruction in the higher altitudes.

**Student Activity**

Mark the direction of the primary winds in the given world map.




Not to scale



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